

Hospital Patient Referral, Treatment and Administration System

UFCEP4-0-3 Business Process Modelling and Enterprise Architecture

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Project Aim and Requirement Basis

Business process design for the hospital patient pathway

Support the pathway from referral through appointment, treatment, follow-up and ongoing communication.
Improve coordination between clinical, administrative and financial teams.
Keep clinical decisions, administrative tasks and financial responsibilities separate.
Provide a basis for acceptance criteria and later testing.

Requirements basis

- Functional requirements
- Non-functional requirements
- Business rules
- Exceptions
- Clarification questions

Team Ownership

First Owner leads; Second Owner supports handover

DENG JIN WANG → Referral & Consultant Review | Second: WU JIA WEI

WU JIA WEI → Appointment & Patient Communication | Second: LIN JIA QIANG

LIN JIA QIANG → Treatment & Chemotherapy | Second: WU TONG YU

WU TONG YU → Funding / Payment / Refund | Second: XIAO YU YING

XIAO YU YING → Enquiry / Clinic Letter / Pathway Monitoring | Second: DENG JIN WANG

Purpose: balanced ownership, clear accountability and continuity if a member is unavailable.

Operational BPMN: End-to-End Model

One integrated process split into five connected areas



Start: Referral received by hospital.

Core handovers: Accepted Referral → Confirmed Appointment → Treatment Status → Financial Status → Correspondence/Monitoring.

The editable BPMN file contains participants, activities, gateways, messages and exception paths.

Referral and Appointment Logic

DENG JIN WANG + WU JIA WEI

Referral & Consultant Review

- Receive and record referral
- Check supporting information
- Request missing information where required
- Send checked referral to Consultant
- Consultant decides: accept, reject, request further information or redirect
- Record decision, reason and decision maker

Appointment & Patient Communication

- Starts only after accepted referral
- Receive booking requirements
- Search external scheduling service
- Confirm appointment in hospital system
- Send appointment letter
- If within two weeks, also telephone patient
- Record contact outcome and exceptions

Treatment and Finance Logic

LIN JIA QIANG + WU TONG YU

Treatment & Chemotherapy

- Patient assessment and treatment discussion
- Record consent
- Create authorised Treatment Booking Request
- Reject/return unauthorised requests
- Check treatment/lab/imaging availability
- Keep pending if external service unavailable
- Continue, delay or modify after clinical review

Funding / Payment / Refund

- Determine funding source
- Record insurer/funding approval if required
- Send secure payment request
- Record payment result
- Handle declined, cancelled, incomplete or duplicate payment
- Investigate lost confirmation
- Process authorised refunds

Enquiry, Clinic Letter and Monitoring Logic

XIAO YU YING

Every enquiry is recorded, classified, prioritised and routed.

Administrative enquiries may be handled by call handlers within authorised information.

Financial enquiries are referred to Finance when required.

Clinical enquiries go to qualified clinical staff; call handlers must not give clinical advice.

Consultant prepares and approves Clinic Letter before distribution.

Delayed letters enter pathway monitoring: weekly reminders, one-month escalation, three-month escalation.

System Boundary and External Interactions

External organisations/services are outside the hospital pool

External services / organisations

Referring Organisation: sends referral and supporting information; receives missing-information request

External Scheduling Service: provides appointment availability

External Correspondence Service: sends appointment letters and Clinic Letters

External Treatment/Lab/Imaging Services: provides treatment resource availability

External Payment Service Provider: processes payment/refund requests

The hospital system records requests, decisions, outcomes and audit evidence; external services return service responses.

Acceptance Test Plan

How acceptance criteria are structured

Each criterion has a unique ID, e.g. AC-06.
Each criterion links to a requirement or business rule.
Each test includes scenario, preconditions, test data, actions, expected result and pass/fail condition.
Coverage includes normal workflow, alternatives, exceptions and non-functional expectations.

Coverage

20 acceptance criteria across all five owner areas, including normal, alternative, exception and non-functional requirements.

Selected Acceptance Criteria

Examples from the submitted plan

AC-03: Consultant decision routes correctly and records reason + decision maker

AC-06: If appointment is within two weeks, telephone contact is required in addition to letter

AC-10: If external treatment service is unavailable, booking remains pending and duplicate appointment is not created

AC-14: If payment confirmation is lost, transaction is investigated and patient is not automatically charged again

AC-18: Delayed Clinic Letter receives reminders and one-month / three-month escalation

AC-19 / AC-20: Role-based access and audit trail must work

Justification of Model Decisions

Why the model is designed this way

Decision justification
Five areas: matches the patient pathway and provides balanced team ownership
Referral separated from appointment: booking only starts after Consultant acceptance
Role boundaries: clinical, administrative and financial decisions are separated
External services: scheduling, correspondence, treatment resources and payment provider sit outside hospital boundary
Exception gateways: missing info, no slot, payment failure and delayed letters require different outcomes
Assumptions recorded: urgency rules, contact attempts and service retry rules need clarification

Assumptions and Clarifications

Open questions to confirm with tutor/client

Exact criteria for urgent referrals are not defined.
Exact criteria for urgent clinical enquiries are not defined.
Number of telephone contact attempts before escalation is not specified.
Retry/timeout rules for unavailable external services are not specified.
Actions after cancellation, decline or failure to attend require clarification.

These assumptions are documented and should be updated after tutor feedback.